

Hospital Emergency Room Triage System — Abstract

The Hospital Emergency Room Triage System is a software application designed to streamline and automate the initial patient assessment, prioritization, and assignment process in a hospital emergency department. Traditional paper-based triage methods often lead to delays, human error, and difficulty in tracking patient urgency, which can compromise critical care delivery. This system provides a centralized platform to efficiently classify patients based on medical severity and manage emergency room workflows.

The system enables medical staff to register incoming patients, record vital signs, assign triage priority levels (e.g., Immediate, Urgent, Non-Urgent), and track waiting times. It maintains real-time records of patient status, bed availability, and attending staff assignments. Healthcare providers can search for patient records, update medical statuses, and identify critical cases requiring immediate intervention.

By automating routine intake and prioritizing care based on medical necessity, the proposed system reduces admission delays, minimizes manual record errors, saves crucial time, and improves patient outcome tracking. It provides an organized, reliable method for monitoring emergency room capacity and generating operational reports. The Hospital Emergency Room Triage System can be implemented as a Java application and further enhanced with features like real-time vital monitor integration, automated SMS alerts for family members, queue time estimation algorithms, and role-based access for doctors and nurses. Overall, the system aims to deliver a fast, structured, and user-friendly solution for managing emergency care activities.